

# ISC Class XII Physics

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## Chapter 1 Electric Charges and Fields

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### 39 Numericals Answer Key

Step-by-Step ISC Board Solutions

#### Topics Covered

- I. Electric Charge & Quantisation
- II. Coulomb's Law & Superposition
- III. Equilibrium of Charges
- IV. Electric Field Lines & Intensity
- V. Electric Dipole
- VI. Electric Flux (Gauss's Law)
- VII. Linear Charge Distribution
- VIII. Sheet(s) of Charge
- IX. Charged Spherical Shell

| Q.No. | Topic / Question Summary                                            | Section |
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| 1     | Number of electrons for 1 C charge                                  | I       |
| 2     | Charge & mass change when electrons removed from glass rod          | I       |
| 3     | Conservation of charge in nuclear reaction                          | I       |
| 4     | Coulombian force between proton and electron                        | II      |
| 5     | Two positive charges: sum given, find individual charges            | II      |
| 6     | Dust particles — number of electrons on each                        | II      |
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| 9     | Distance between electrons/protons when $F = mg$                    | II      |
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| 16    | Charge at centre of square for system equilibrium                   | III     |
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| 28    | Electric flux through cubical surface due to point charge           | VI      |
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| 30    | Flux through doubled Gaussian surface; magnitude of charge          | VI      |
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**Verification note:** All 39 answers have been cross-checked against the *Nootan ISC Physics-XII* textbook answer key. Two issues found and corrected from the original AI-generated draft: **Q8** — the charge formula was mislabelled ( $n^3$ , not  $n/3$ , though the calculation was correct); **Q13** — the step-by-step geometric explanation was wrong (incorrectly claimed the pair-force was  $9\sqrt{3} \times 10^{-3}$  N along AD, then jumped to the answer); the corrected solution uses component resolution and shows correctly that the two perpendicular  $9 \times 10^{-3}$  N forces give  $9\sqrt{2} \times 10^{-3}$  N at  $45^\circ$ .

## Formula Sheet

### I. Electric Charge and Quantisation

**Quantisation of charge:**  $Q = ne$

$n$  = number of electrons,  $e = 1.6 \times 10^{-19}$  C

### II. Coulomb's Law and Superposition

**Coulomb's Law:**  $F = k \frac{q_1 q_2}{r^2}$

$k = 9 \times 10^9$  N m<sup>2</sup>C<sup>-2</sup>; like charges repel, unlike attract

**Force due to  $g$ :**  $F = k \frac{e^2}{r^2} = mg$

used to find separation at which electrostatic force equals weight

**Superposition principle:**  $\vec{F}_{net} = \sum \vec{F}_i$

vector sum of individual Coulomb forces on a charge

### III. Equilibrium of Charges

**Null point condition:**  $k \frac{q_1 q_3}{r_1^2} = k \frac{q_2 q_3}{r_2^2}$

net force on  $q_3$  is zero; null point lies on the side of the smaller charge if signs are opposite, and between the charges if signs are the same

### IV. Electric Field Lines and Intensity

**Electric field (point charge):**  $E = k \frac{q}{r^2}$

field points away from  $+q$ , towards  $-q$

**Force on a charge in a field:**  $F = qE$

direction along  $E$  for  $+q$ , opposite for  $-q$

**Field in vector form:**  $\vec{E} = k \frac{q}{r^3} \vec{r}$

$\vec{r}$  = position vector from source charge to field point

### V. Electric Dipole

**Dipole moment:**  $p = q \times 2l$

$2l$  = separation between the two charges

**Axial (end-on) field:**  $E_{\parallel} = \frac{2kp}{r^3}$

valid for  $r \gg l$

**Equatorial (broadside-on) field:**  $E_{\perp} = \frac{kp}{r^3}$

valid for  $r \gg l$

**Torque on a dipole:**  $\tau = pE \sin \theta$

maximum at  $\theta = 90^\circ$ :  $\tau_{max} = pE$

## VI. Electric Flux and Gauss's Law

**Electric flux:**  $\phi = \vec{E} \cdot \vec{A} = EA \cos \theta$

$\theta$  = angle between  $\vec{E}$  and the outward normal  $\hat{n}$

**Gauss's Law:**  $\phi = \frac{q_{enc}}{\epsilon_0}$

flux depends only on charge enclosed, not on the size/shape of the surface

## VII. Linear Charge Distribution

**Field due to infinite line charge:**  $E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{2k\lambda}{r}$

$\lambda$  = linear charge density (C/m)

## VIII. Sheet(s) of Charge

**Field due to a charged sheet:**  $E = \frac{\sigma}{2\epsilon_0}$

field due to one plate;  $\sigma$  = surface charge density

**Field between oppositely charged plates:**  $E = \frac{\sigma}{\epsilon_0}$

fields of both plates add between them and cancel outside

## IX. Charged Spherical Shell

**Charge from surface density:**  $Q = 4\pi R^2 \sigma$

$R$  = radius of sphere

**Field of a charged conducting sphere:**  $E_{in} = 0$ ;  $E_{surface} = k \frac{Q}{R^2}$ ;  $E_{out} = k \frac{Q}{r^2}$

all charge behaves as if concentrated at the centre for points outside

## I. Electric Charge and its Quantisation

### Question 1

Calculate the number of electrons to have a total negative charge of 1 C.

**Given:** Total charge  $Q = 1 \text{ C}$ ;  $e = 1.6 \times 10^{-19} \text{ C}$

**Formula:**  $Q = ne \implies n = \frac{Q}{e}$

**Solution:**

$$n = \frac{Q}{e} = \frac{1}{1.6 \times 10^{-19}} = 6.25 \times 10^{18}$$

### Final Answer

$$n = 6.25 \times 10^{18} \text{ electrons}$$

**Common Mistake:** Do not confuse the charge of one electron ( $1.6 \times 10^{-19} \text{ C}$ ) with the charge of one mole of electrons (Faraday = 96500 C).

### Question 2

$6.25 \times 10^{12}$  electrons are removed from a glass rod.

- (i) What is the magnitude and nature of charge acquired by the rod?
- (ii) What will be the change in mass of the rod? (mass of electron =  $9.0 \times 10^{-31} \text{ kg}$ )

**Given:**  $n = 6.25 \times 10^{12}$ ;  $e = 1.6 \times 10^{-19} \text{ C}$ ;  $m_e = 9.0 \times 10^{-31} \text{ kg}$

**Formula:**  $Q = ne$ ;  $\Delta m = n \cdot m_e$

**Solution:**

(i)

$$Q = ne = 6.25 \times 10^{12} \times 1.6 \times 10^{-19} = 1.0 \times 10^{-6} \text{ C} = 1.0 \mu\text{C}$$

Since electrons (negative charges) are *removed*, the rod acquires a **positive** charge.

(ii)

$$\Delta m = n \cdot m_e = 6.25 \times 10^{12} \times 9.0 \times 10^{-31} = 5.625 \times 10^{-18} \text{ kg}$$

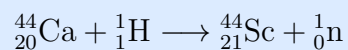
The mass of the rod **decreases** by  $5.625 \times 10^{-18} \text{ kg}$ .

### Final Answer

- (i)  $1.0 \mu\text{C}$ , positive; (ii) Mass decreases by  $5.625 \times 10^{-18} \text{ kg}$

**Question 3**

Is the law of conservation of charge valid in the reaction:



**Solution:**

- **Left side:** Atomic number of Ca = 20, of H = 1. Total charge =  $20 + 1 = 21$ .
- **Right side:** Atomic number of Sc = 21, of neutron = 0. Total charge =  $21 + 0 = 21$ .

Both sides have the same total charge (21 units). Hence the law of conservation of charge is **valid**.

**Final Answer**

Yes, the law of conservation of charge is obeyed. ( $21 = 21$ )

## II. Coulomb's Law and Superposition Principle

### Question 4

Calculate the Coulombian force between a proton and an electron separated by  $0.8 \times 10^{-15}$  m.

**Given:**  $q_1 = q_2 = 1.6 \times 10^{-19}$  C;  $r = 0.8 \times 10^{-15}$  m;  $k = 9 \times 10^9$  N m<sup>2</sup>C<sup>-2</sup>

**Formula:**  $F = k \frac{q_1 q_2}{r^2}$

**Solution:**

$$F = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{(0.8 \times 10^{-15})^2} = \frac{9 \times 10^9 \times 2.56 \times 10^{-38}}{0.64 \times 10^{-30}} = \frac{2.304 \times 10^{-28}}{6.4 \times 10^{-31}} = 360 \text{ N}$$

The force is **attractive** (unlike charges).

### Final Answer

$F = 360$  N (attractive)

### ★ Likely Exam Question

### Question 5

Two positive charges 0.1 m apart repel each other with a force of 18 N. If the sum of charges is  $9 \mu\text{C}$ , find their separate values.

**Given:**  $r = 0.1$  m;  $F = 18$  N;  $q_1 + q_2 = 9 \times 10^{-6}$  C;  $k = 9 \times 10^9$  N m<sup>2</sup>C<sup>-2</sup>

**Formula:**  $F = k \frac{q_1 q_2}{r^2}$

**Solution:**

$$q_1 q_2 = \frac{F r^2}{k} = \frac{18 \times (0.1)^2}{9 \times 10^9} = \frac{18 \times 10^{-2}}{9 \times 10^9} = 2 \times 10^{-11} \text{ C}^2 = 20 \times 10^{-12} \text{ C}^2$$

Let  $q_1 = x$ ,  $q_2 = (9 - x) \times 10^{-6}$ . Then:

$$x(9 - x) \times 10^{-12} = 20 \times 10^{-12} \implies x^2 - 9x + 20 = 0 \implies (x - 5)(x - 4) = 0$$

So  $x = 5$  or  $x = 4$ , i.e.  $q_1 = 5 \mu\text{C}$  and  $q_2 = 4 \mu\text{C}$ .

### Final Answer

$q_1 = 5 \mu\text{C}$ ,  $q_2 = 4 \mu\text{C}$

**Question 6**

Two negatively-charged dust particles are 1 mm apart and repel with  $4.0 \times 10^{-5}$  N. Charge on one is 4 times the other. Find the number of electrons on each. (Charge on one electron =  $\frac{5}{3} \times 10^{-19}$  C)

**Given:**  $r = 1 \times 10^{-3}$  m;  $F = 4.0 \times 10^{-5}$  N;  $q_2 = 4q_1$ ;  $e = \frac{5}{3} \times 10^{-19}$  C

**Formula:**  $F = k \frac{q_1 q_2}{r^2}$ ;  $n = \frac{q}{e}$

**Solution:**

$$F = k \frac{q_1 \cdot 4q_1}{r^2} \implies 4 \times 10^{-5} = \frac{9 \times 10^9 \times 4q_1^2}{(10^{-3})^2} = \frac{36 \times 10^9 q_1^2}{10^{-6}}$$

$$q_1^2 = \frac{4 \times 10^{-5} \times 10^{-6}}{36 \times 10^9} = \frac{4 \times 10^{-11}}{36 \times 10^9} = \frac{1}{9} \times 10^{-20} \implies q_1 = \frac{1}{3} \times 10^{-10} \text{ C}$$

$$n_1 = \frac{q_1}{e} = \frac{\frac{1}{3} \times 10^{-10}}{\frac{5}{3} \times 10^{-19}} = \frac{10^{-10}}{5 \times 10^{-19}} = 2 \times 10^8$$

$$q_2 = 4q_1 = \frac{4}{3} \times 10^{-10} \text{ C}, \quad n_2 = 4n_1 = 8 \times 10^8$$

**Final Answer**

$n_1 = 2 \times 10^8$  electrons;  $n_2 = 8 \times 10^8$  electrons

**Question 7**

$+2 \mu\text{C}$  and  $+6 \mu\text{C}$  repel each other with 12 N. Each is given an additional charge of  $-4 \mu\text{C}$ . What is the new force?

**Given:**  $q_1 = +2 \mu\text{C}$ ,  $q_2 = +6 \mu\text{C}$ ,  $F_1 = 12$  N, additional charge =  $-4 \mu\text{C}$

**Formula:**  $F = k \frac{q_1 q_2}{r^2}$

**Solution:**

New charges:  $q'_1 = 2 - 4 = -2 \mu\text{C}$ ;  $q'_2 = 6 - 4 = +2 \mu\text{C}$ .

From initial condition:

$$r^2 = \frac{k q_1 q_2}{F_1} = \frac{9 \times 10^9 \times 2 \times 10^{-6} \times 6 \times 10^{-6}}{12} = \frac{108 \times 10^{-3}}{12} = 9 \times 10^{-3} \text{ m}^2$$

New force:

$$F_2 = k \frac{|q'_1| |q'_2|}{r^2} = \frac{9 \times 10^9 \times 2 \times 10^{-6} \times 2 \times 10^{-6}}{9 \times 10^{-3}} = \frac{36 \times 10^{-3}}{9 \times 10^{-3}} = 4 \text{ N}$$

Since the charges are opposite in sign, the force is **attractive**.

**Final Answer**

$$F_2 = 4 \text{ N (attraction)}$$

**Question 8**

Ten positively charged particles at  $x = 10, 20, \dots, 100$  cm carry charges  $n^3 \times 10^{-8}$  C for  $n = 1, 2, \dots, 10$  (so the charges are  $1 \times 10^{-8}, 8 \times 10^{-8}, 27 \times 10^{-8}, \dots, 1000 \times 10^{-8}$  C). Find the net electric force on a 1 C charge at the origin.

**Given:**  $q_n = n^3 \times 10^{-8}$  C at  $x_n = 0.1n$  m; Test charge  $Q = 1$  C at origin;  $k = 9 \times 10^9 \text{ N m}^2\text{C}^{-2}$

**Formula:**  $F = kQ \sum_{n=1}^{10} \frac{q_n}{x_n^2}$

**Solution:**

Each force is repulsive (all positive). The  $n$ -th force:

$$F_n = k \cdot 1 \cdot \frac{n^3 \times 10^{-8}}{(0.1n)^2} = 9 \times 10^9 \times \frac{n^3 \times 10^{-8}}{0.01n^2} = 9 \times 10^9 \times 10^{-6} \times n$$

Summing over  $n = 1$  to 10:

$$\sum_{n=1}^{10} n = \frac{10 \times 11}{2} = 55$$

$$F = 9 \times 10^9 \times 10^{-6} \times 55 = 9 \times 10^3 \times 55 = 4.95 \times 10^5 \text{ N}$$

All forces point in the  $-x$  direction (away from positive particles toward origin).

**Final Answer**

$$F = 4.95 \times 10^5 \text{ N (directed towards } -x \text{ axis)}$$

**★ Likely Exam Question****Question 9**

How far apart should two electrons be so that the force between them equals the weight of an electron? What about protons?

**Given:**  $e = 1.6 \times 10^{-19}$  C;  $m_e = 9.1 \times 10^{-31}$  kg;  $m_p = 1.67 \times 10^{-27}$  kg;  $g = 9.8 \text{ m/s}^2$ ;  $k = 9 \times 10^9 \text{ N m}^2\text{C}^{-2}$

**Formula:**  $F = k \frac{e^2}{r^2} = mg \implies r = e \sqrt{\frac{k}{mg}}$

**Solution:**

**Electrons:**

$$r = e\sqrt{\frac{k}{m_e g}} = 1.6 \times 10^{-19} \sqrt{\frac{9 \times 10^9}{9.1 \times 10^{-31} \times 9.8}} = 1.6 \times 10^{-19} \sqrt{\frac{9 \times 10^9}{8.918 \times 10^{-30}}}$$

$$= 1.6 \times 10^{-19} \sqrt{1.009 \times 10^{39}} = 1.6 \times 10^{-19} \times 1.0045 \times 10^{19.5} \approx 1.6 \times 10^{-19} \times 3.18 \times 10^{19} \approx 5.1 \text{ m}$$

**Protons:**

$$r = 1.6 \times 10^{-19} \sqrt{\frac{9 \times 10^9}{1.67 \times 10^{-27} \times 9.8}} = 1.6 \times 10^{-19} \sqrt{\frac{9 \times 10^9}{1.637 \times 10^{-26}}}$$

$$= 1.6 \times 10^{-19} \sqrt{5.498 \times 10^{35}} = 1.6 \times 10^{-19} \times 7.415 \times 10^{17} \approx 0.117 \text{ m}$$

### Final Answer

Electrons:  $r \approx 5.1 \text{ m}$ ; Protons:  $r \approx 0.117 \text{ m}$

### Question 10

Balls A ( $+40 \mu\text{C}$ ) and B ( $-10 \mu\text{C}$ ) are 2.0 m apart.

(a) Force between them. (b) They touch; force after separation. (c) One ball grounded.

**Given:**  $q_A = +40 \mu\text{C}$ ,  $q_B = -10 \mu\text{C}$ ,  $r = 2.0 \text{ m}$

**Formula:**  $F = k \frac{|q_A||q_B|}{r^2}$

**Solution:**

(a) **Initial force:**

$$F_1 = \frac{9 \times 10^9 \times 40 \times 10^{-6} \times 10 \times 10^{-6}}{(2)^2} = \frac{9 \times 10^9 \times 4 \times 10^{-10}}{4} = 0.90 \text{ N}$$

Force is **attractive** (unlike charges).

(b) **After touching:** charge redistributes equally:

$$q' = \frac{q_A + q_B}{2} = \frac{40 - 10}{2} \mu\text{C} = 15 \mu\text{C} \text{ each (positive)}$$

$$F_2 = \frac{9 \times 10^9 \times (15 \times 10^{-6})^2}{(2)^2} = \frac{9 \times 10^9 \times 2.25 \times 10^{-10}}{4} = 0.506 \approx 0.51 \text{ N}$$

Force is **repulsive**.

(c) **If one ball is connected to earth:** it discharges to zero. So  $F = 0$ .

### Final Answer

(a) 0.90 N (attraction); (b) 0.51 N (repulsion); (c) Zero

**Question 11**

Spheres A and B (charge  $+q$  each) repel with  $F_0 = 2.0 \times 10^{-5}$  N, separated by distance  $d$ . Uncharged sphere C touches A, then placed at mid-point. Find net force on C.

**Formula:**  $F = k \frac{q_1 q_2}{r^2}$

**Solution:**

After C touches A: A and C share charge equally  $\Rightarrow$  each gets  $q/2$ .

Let separation AB =  $d$ . Mid-point of AB is at distance  $d/2$  from each.

**Force on C from A:**

$$F_{CA} = k \frac{(q/2)(q/2)}{(d/2)^2} = k \frac{q^2/4}{d^2/4} = k \frac{q^2}{d^2} = F_0 = 2.0 \times 10^{-5} \text{ N}$$

(repulsive, directed away from A, i.e. towards B)

**Force on C from B:**

$$F_{CB} = k \frac{(q/2)(q)}{(d/2)^2} = k \frac{q^2/2}{d^2/4} = 2k \frac{q^2}{d^2} = 2F_0 = 4.0 \times 10^{-5} \text{ N}$$

(repulsive, directed away from B, i.e. towards A)

**Net force on C:**

$$F_{net} = F_{CB} - F_{CA} = 4.0 \times 10^{-5} - 2.0 \times 10^{-5} = 2.0 \times 10^{-5} \text{ N}$$

Direction: **towards A.**

**Final Answer**

$$F_{net} = 2.0 \times 10^{-5} \text{ N, directed towards A}$$

**Question 12**

A charge  $2.0 \times 10^{-4}$  C is placed directly below and 10 cm from the bob of a simple pendulum of mass 100 g. What charge should the bob have so that the string becomes loose?

**Given:**  $q_1 = 2.0 \times 10^{-4}$  C;  $r = 0.10$  m;  $m = 0.1$  kg;  $g = 9.8$  m/s<sup>2</sup>

**Formula:** String becomes loose when upward electric force =  $mg$

**Solution:**

String becomes loose when:

$$F_e = mg \implies k \frac{q_1 q_2}{r^2} = mg$$

$$q_2 = \frac{mgr^2}{kq_1} = \frac{0.1 \times 9.8 \times (0.1)^2}{9 \times 10^9 \times 2.0 \times 10^{-4}} = \frac{0.1 \times 9.8 \times 10^{-2}}{18 \times 10^5} = \frac{9.8 \times 10^{-3}}{18 \times 10^5} = 5.44 \times 10^{-9} \text{ C} \approx 5.4 \text{ nC}$$

Charge must be **same sign** as  $q_1$  (both positive  $\Rightarrow$  repulsion, i.e. upward force on bob).

### Final Answer

$$q_2 = 5.4 \times 10^{-9} \text{ C (same sign as } q_1)$$

### ★ Likely Exam Question

### Question 13

In equilateral triangle ABC (side 10 m), D is mid-point of BC. Charges:  $+100 \mu\text{C}$  at B,  $-100 \mu\text{C}$  at C,  $+75 \mu\text{C}$  at D. Find force on  $+1 \mu\text{C}$  at A.

**Given:** Side  $a = 10 \text{ m}$ ;  $q_B = 100 \mu\text{C}$ ,  $q_C = -100 \mu\text{C}$ ,  $q_D = 75 \mu\text{C}$ ,  $q_A = 1 \mu\text{C}$

**Formula:** Coulomb's law + vector superposition

### Solution:

Set up coordinates with D at the origin, B at  $(-5, 0)$ , C at  $(5, 0)$ , and A at  $(0, 5\sqrt{3}) \text{ m}$ .

$$k = 9 \times 10^9 \text{ N m}^2\text{C}^{-2}, q_A = 1 \times 10^{-6} \text{ C}.$$

**Forces due to  $q_B$  and  $q_C$ :**

$$|F_{AB}| = |F_{AC}| = \frac{9 \times 10^9 \times 100 \times 10^{-6} \times 1 \times 10^{-6}}{(10)^2} = 9 \times 10^{-3} \text{ N}$$

$F_{AB}$  acts along BA direction (repulsion from B);  $F_{AC}$  acts along AC direction (attraction toward C). Resolving into components:

$$F_{AB,x} = 9 \times 10^{-3} \sin 30^\circ = 4.5 \times 10^{-3} \text{ N} \quad (+x)$$

$$F_{AC,x} = 9 \times 10^{-3} \sin 30^\circ = 4.5 \times 10^{-3} \text{ N} \quad (+x)$$

The  $y$ -components cancel (equal and opposite). Hence the resultant of  $F_{AB}$  and  $F_{AC}$ :

$$\vec{F}_{BC\text{-pair}} = (9 \times 10^{-3} \text{ N}) \hat{x} \quad (\text{parallel to BC})$$

**Force due to  $q_D$  at A ( $AD = 5\sqrt{3} \text{ m}$ ):**

$$F_{AD} = \frac{9 \times 10^9 \times 75 \times 10^{-6} \times 1 \times 10^{-6}}{(5\sqrt{3})^2} = \frac{9 \times 10^9 \times 75 \times 10^{-12}}{75} = 9 \times 10^{-3} \text{ N}$$

Direction: repulsion from D — *upward* along  $+y$  (away from D, i.e. away from BC).

**Net force:** The two components are *perpendicular*:

$$F_{\text{net}} = \sqrt{(9 \times 10^{-3})^2 + (9 \times 10^{-3})^2} = 9\sqrt{2} \times 10^{-3} \text{ N}$$

Direction: at  $45^\circ$  to the line BC (bisecting the angle between the  $+x$  and  $+y$  directions).

### Final Answer

$$F = 9\sqrt{2} \times 10^{-3} \text{ N, at } 45^\circ \text{ to a line parallel to BC}$$

### III. Equilibrium of a Charge and System of Charges

#### Question 14

Charges  $+1.0 \mu\text{C}$  and  $-0.25 \mu\text{C}$  are  $0.40 \text{ m}$  apart. Where should a third charge be placed so that no force acts on it?

**Given:**  $q_1 = +1.0 \mu\text{C}$ ,  $q_2 = -0.25 \mu\text{C}$ ,  $d = 0.40 \text{ m}$

#### Solution:

For equilibrium, the third charge must be on the line joining the two charges. Since the charges are **opposite in sign**, the neutral point lies **outside** the line segment, on the side of the *smaller* charge (i.e. beyond  $q_2$ ).

Let the distance from  $q_2$  be  $x$ . Then distance from  $q_1 = d + x = 0.40 + x$ .

$$\frac{kq_1q_3}{(0.40 + x)^2} = \frac{k|q_2|q_3}{x^2} \implies \frac{1.0}{(0.40 + x)^2} = \frac{0.25}{x^2}$$

$$\frac{x^2}{(0.40 + x)^2} = 0.25 \implies \frac{x}{0.40 + x} = 0.5 \implies 2x = 0.40 + x \implies x = 0.40 \text{ m}$$

#### Final Answer

The third charge should be placed at  $0.40 \text{ m}$  beyond  $q_2$ , away from  $q_1$ .

#### ★ Likely Exam Question

#### Question 15

Three charges  $q_1$ ,  $q_2$ ,  $q_3$  are collinear and equally spaced.  $q_2$  and  $q_3$  are opposite in sign. Find  $q_1$  such that net force on  $q_3 = 0$ .

#### Solution:

Let separation =  $d$ . Forces on  $q_3$ :

- From  $q_2$  (distance  $d$ ):  $F_{23} = k \frac{|q_2||q_3|}{d^2}$
- From  $q_1$  (distance  $2d$ ):  $F_{13} = k \frac{|q_1||q_3|}{4d^2}$

For  $q_3$  to be in equilibrium,  $F_{13} = F_{23}$  and opposite direction:

$$\frac{k|q_1||q_3|}{4d^2} = \frac{k|q_2||q_3|}{d^2} \implies |q_1| = 4|q_2|$$

Since  $q_2$  and  $q_3$  are opposite in sign, the force from  $q_2$  on  $q_3$  is attractive. For  $q_1$  to repel  $q_3$ ,  $q_1$  and  $q_3$  must have the same sign.

#### Final Answer

$q_1$  has magnitude  $4|q_2|$  and the same sign as  $q_3$ .

**Question 16**

At each corner of a square of side  $a$ , a charge  $+q$  is placed. What charge at the centre ensures equilibrium for the whole system?

**Solution:**

Consider one corner charge (say at A). Forces on it:

- From each adjacent corner (distance  $a$ ):  $F_1 = k \frac{q^2}{a^2}$ , along the diagonal outward.
- From the diagonally opposite corner (distance  $a\sqrt{2}$ ):  $F_2 = k \frac{q^2}{2a^2}$ .
- From centre charge  $Q$  (distance  $\frac{a\sqrt{2}}{2} = \frac{a}{\sqrt{2}}$ ):  $F_Q = k \frac{qQ}{a^2/2} = \frac{2kqQ}{a^2}$ .

Net outward force from the three corner charges (along diagonal):

$$F_{out} = 2F_1 \cos 45^\circ + F_2 = 2 \cdot k \frac{q^2}{a^2} \cdot \frac{1}{\sqrt{2}} + k \frac{q^2}{2a^2} = \frac{kq^2}{a^2} \left( \sqrt{2} + \frac{1}{2} \right) = \frac{kq^2}{a^2} \cdot \frac{2\sqrt{2} + 1}{2}$$

For equilibrium,  $F_Q$  (inward) =  $F_{out}$ :

$$\frac{2kqQ}{a^2} = \frac{kq^2(2\sqrt{2} + 1)}{2a^2} \implies Q = \frac{q(2\sqrt{2} + 1)}{4} \times (-1) = -\frac{q(1 + 2\sqrt{2})}{4}$$

**Final Answer**

$$Q = -\frac{q(1 + 2\sqrt{2})}{4}$$

**Question 17**

$+9e$  and  $+e$  are 16 cm apart. Where should a third charge  $q$  be placed for equilibrium?

**Given:**  $q_1 = 9e$ ,  $q_2 = e$ ,  $d = 16 \text{ cm} = 0.16 \text{ m}$

**Solution:**

Let the third charge be at distance  $x$  from  $q_1$  and  $(16 - x)$  cm from  $q_2$ .

Both charges are positive; the neutral point is **between** them.

$$k \frac{9e \cdot q}{x^2} = k \frac{e \cdot q}{(16 - x)^2} \implies \frac{9}{x^2} = \frac{1}{(16 - x)^2} \implies 3(16 - x) = x \implies 48 - 3x = x \implies x = 12 \text{ cm}$$

**Final Answer**

The third charge should be placed 12 cm from  $q_1$  ( $+9e$ ), i.e. 4 cm from  $q_2$  ( $+e$ ).

## IV. Electric Field Lines and Intensity

### Question 18

Find the electric field due to a helium nucleus (charge =  $2e$ ) at a distance of  $1 \text{ \AA}$ .

**Given:**  $q = 2e = 2 \times 1.6 \times 10^{-19} = 3.2 \times 10^{-19} \text{ C}$ ;  $r = 1 \times 10^{-10} \text{ m}$

**Formula:**  $E = k \frac{q}{r^2}$

**Solution:**

$$E = \frac{9 \times 10^9 \times 3.2 \times 10^{-19}}{(10^{-10})^2} = \frac{2.88 \times 10^{-9}}{10^{-20}} = 2.88 \times 10^{11} \text{ N C}^{-1}$$

### Final Answer

$$E = 2.88 \times 10^{11} \text{ N C}^{-1}$$

### Question 19

$q_1 = +0.2 \text{ C}$  and  $q_2 = +0.4 \text{ C}$  are  $0.1 \text{ m}$  apart. Find  $E$  at:

(i) midpoint; (ii) a point  $0.05 \text{ m}$  from  $q_2$  and  $0.15 \text{ m}$  from  $q_1$ .

**Given:**  $q_1 = 0.2 \text{ C}$ ,  $q_2 = 0.4 \text{ C}$ ,  $d = 0.1 \text{ m}$

**Formula:**  $E = k \frac{q}{r^2}$  (fields point away from positive charges)

**Solution:**

(i) Midpoint ( $r = 0.05 \text{ m}$  from each):

$$E_1 = k \frac{q_1}{(0.05)^2} = \frac{9 \times 10^9 \times 0.2}{2.5 \times 10^{-3}} = 7.2 \times 10^{11} \text{ N C}^{-1} \text{ (towards } q_2)$$

$$E_2 = k \frac{q_2}{(0.05)^2} = \frac{9 \times 10^9 \times 0.4}{2.5 \times 10^{-3}} = 1.44 \times 10^{12} \text{ N C}^{-1} \text{ (towards } q_1)$$

$$E_{\text{net}} = E_2 - E_1 = (14.4 - 7.2) \times 10^{11} = 7.20 \times 10^{11} \text{ N C}^{-1} \text{ towards } q_1$$

(ii) Point  $0.15 \text{ m}$  from  $q_1$ ,  $0.05 \text{ m}$  from  $q_2$  (outside, beyond  $q_2$ ):

$$E'_1 = k \frac{0.2}{(0.15)^2} = \frac{9 \times 10^9 \times 0.2}{0.0225} = 8 \times 10^{10} \text{ N C}^{-1} \text{ (away from } q_1)$$

$$E'_2 = k \frac{0.4}{(0.05)^2} = \frac{9 \times 10^9 \times 0.4}{2.5 \times 10^{-3}} = 1.44 \times 10^{12} \text{ N C}^{-1} \text{ (away from } q_2)$$

Both point in same direction (away from both  $q_1$  and  $q_2$ , since point is beyond  $q_2$ ):

$$E_{\text{net}} = E'_1 + E'_2 \approx 0.08 \times 10^{12} + 1.44 \times 10^{12} = 1.52 \times 10^{12} \text{ N C}^{-1}$$

**Final Answer**

(i)  $7.20 \times 10^{11} \text{ N C}^{-1}$  towards  $q_1$ ; (ii)  $1.52 \times 10^{12} \text{ N C}^{-1}$  away from both charges

**Question 20**

Charges  $+1000 \mu\text{C}$  at A and  $-1000 \mu\text{C}$  at B, 2 m apart. Find  $E$  at:

(i) midpoint of AB; (ii) a point equidistant (4 m) from each charge.

**Given:**  $|q| = 1000 \mu\text{C} = 10^{-3} \text{ C}$ ;  $d_{AB} = 2 \text{ m}$

**Solution:**

(i) **Midpoint:**  $r = 1 \text{ m}$  from each.

$$E_A = k \frac{10^{-3}}{1^2} = 9 \times 10^6 \text{ N C}^{-1} \text{ (away from A, towards midpoint)}$$

$$E_B = k \frac{10^{-3}}{1^2} = 9 \times 10^6 \text{ N C}^{-1} \text{ (towards B)}$$

Both fields point in same direction (from A to B):

$$E_{\text{net}} = 9 \times 10^6 + 9 \times 10^6 = 1.8 \times 10^7 \text{ N C}^{-1} \text{ along AB (A to B)}$$

(ii) **Point P equidistant (4 m) from each:** P is above the midpoint of AB.

$AP = BP = 4 \text{ m}$ . The distance from midpoint to P:  $h = \sqrt{16 - 1} = \sqrt{15} \text{ m}$ .

$$E_A = E_B = k \frac{10^{-3}}{16} = \frac{9 \times 10^9 \times 10^{-3}}{16} = 5.625 \times 10^5 \text{ N C}^{-1}$$

The horizontal (along AB) components add; vertical components cancel:

$$\cos \theta = \frac{1}{4} \text{ (half of AB/AP)}$$

$$E_{\text{net}} = 2E_A \cos \theta = 2 \times 5.625 \times 10^5 \times \frac{1}{4} = 2.8125 \times 10^5 \approx 2.8 \times 10^5 \text{ N C}^{-1}$$

Direction: parallel to AB (from A towards B).

**Final Answer**

(i)  $1.8 \times 10^7 \text{ N C}^{-1}$  along AB; (ii)  $2.8 \times 10^5 \text{ N C}^{-1}$  parallel to AB

**Question 21**

A body has an excess of  $10^6$  electrons in an electric field of  $1000 \text{ N/C}$  towards East. Find magnitude and direction of force.

**Given:**  $n = 10^6$ ;  $E = 1000 \text{ N C}^{-1}$  (East);  $e = 1.6 \times 10^{-19} \text{ C}$

**Formula:**  $F = qE = neE$

**Solution:**

$$q = ne = 10^6 \times 1.6 \times 10^{-19} = 1.6 \times 10^{-13} \text{ C}$$

$$F = qE = 1.6 \times 10^{-13} \times 1000 = 1.6 \times 10^{-10} \text{ N}$$

Since the charge is **negative** (excess electrons) and  $E$  points East, force =  $-qE$  points West.

**Final Answer**

$$F = 1.6 \times 10^{-10} \text{ N, directed towards West}$$

**Question 22**

A stationary oil drop has charge  $3.2 \times 10^{-19} \text{ C}$  and weight  $1.6 \times 10^{-14} \text{ N}$ . Find the electric field.

**Given:**  $q = 3.2 \times 10^{-19} \text{ C}$ ;  $W = mg = 1.6 \times 10^{-14} \text{ N}$

**Formula:**  $F_E = qE = W \implies E = \frac{W}{q}$

**Solution:**

$$E = \frac{mg}{q} = \frac{1.6 \times 10^{-14}}{3.2 \times 10^{-19}} = 0.5 \times 10^5 = 5.0 \times 10^4 \text{ N C}^{-1}$$

Direction: upward (balancing gravity).

**Final Answer**

$$E = 5.0 \times 10^4 \text{ N C}^{-1} \text{ (directed upwards)}$$

**Question 23**

A charge of  $2 \mu\text{C}$  is at point  $(1, -2, 1) \text{ m}$ . Find  $\vec{E}$  at  $P(1, 2, 4) \text{ m}$  in vector form.

**Given:**  $q = 2 \times 10^{-6} \text{ C}$ ; Source:  $(1, -2, 1)$ ; Field point:  $(1, 2, 4)$

**Formula:**  $\vec{E} = k \frac{q}{r^3} \vec{r}$

**Solution:**

$$\vec{r} = (1 - 1)\hat{i} + (2 - (-2))\hat{j} + (4 - 1)\hat{k} = 0\hat{i} + 4\hat{j} + 3\hat{k}$$

$$r = \sqrt{0 + 16 + 9} = 5 \text{ m}$$

$$\vec{E} = k \frac{q}{r^3} \vec{r} = \frac{9 \times 10^9 \times 2 \times 10^{-6}}{125} (4\hat{j} + 3\hat{k}) = \frac{18 \times 10^3}{125} (4\hat{j} + 3\hat{k}) = 144(4\hat{j} + 3\hat{k}) \text{ N C}^{-1}$$

**Final Answer**

$$\vec{E} = 144(4\hat{j} + 3\hat{k}) \text{ N C}^{-1}$$

## V. Electric Dipole

### Question 24

Dipole:  $\pm 2.4 \mu\text{C}$ , separation  $2.5 \times 10^{-3} \text{ m}$ . Find  $E$  in broadside-on position at  $0.30 \text{ m}$ . Then dipole rotated  $90^\circ$ ; find new  $E$ .

**Given:**  $q = 2.4 \times 10^{-6} \text{ C}$ ;  $2l = 2.5 \times 10^{-3} \text{ m} \Rightarrow l = 1.25 \times 10^{-3} \text{ m}$ ;  $r = 0.30 \text{ m}$

**Formula:** Broadside-on:  $E_{\perp} = \frac{kp}{(r^2 + l^2)^{3/2}} \approx \frac{kp}{r^3}$  (since  $r \gg l$ )

End-on:  $E_{\parallel} = \frac{2kp}{r^3}$  (since  $r \gg l$ )

**Solution:**

Dipole moment:

$$p = q \times 2l = 2.4 \times 10^{-6} \times 2.5 \times 10^{-3} = 6.0 \times 10^{-9} \text{ C m}$$

**Broadside-on (equatorial) position:**

$$E_{\perp} = \frac{kp}{r^3} = \frac{9 \times 10^9 \times 6 \times 10^{-9}}{(0.3)^3} = \frac{54}{0.027} = 2.0 \times 10^3 \text{ N C}^{-1}$$

**After  $90^\circ$  rotation (End-on/axial):**

$$E_{\parallel} = \frac{2kp}{r^3} = 2 \times 2.0 \times 10^3 = 4.0 \times 10^3 \text{ N C}^{-1}$$

### Final Answer

Broadside-on:  $E = 2.0 \times 10^3 \text{ N C}^{-1}$ ; After  $90^\circ$  (end-on):  $E = 4.0 \times 10^3 \text{ N C}^{-1}$

**Common Mistake:** Broadside-on = equatorial ( $E = kp/r^3$ ); End-on = axial ( $E = 2kp/r^3$ ). Do not mix up.

### Question 25

A dipole at  $30^\circ$  to a uniform field of  $10^4 \text{ N/C}$  experiences torque  $9 \times 10^{-26} \text{ N m}$ . Find dipole moment.

**Given:**  $\theta = 30^\circ$ ;  $E = 10^4 \text{ N C}^{-1}$ ;  $\tau = 9 \times 10^{-26} \text{ N m}$

**Formula:**  $\tau = pE \sin \theta$

**Solution:**

$$p = \frac{\tau}{E \sin \theta} = \frac{9 \times 10^{-26}}{10^4 \times \sin 30^\circ} = \frac{9 \times 10^{-26}}{10^4 \times 0.5} = \frac{9 \times 10^{-26}}{5 \times 10^3} = 1.8 \times 10^{-29} \text{ C m}$$

**Final Answer**

$$p = 1.8 \times 10^{-29} \text{ C m}$$

**Question 26**

Dipole:  $\pm 1 \mu\text{C}$ , separation 2 cm.  $E = 1 \times 10^5 \text{ N C}^{-1}$ ;  $\vec{p}$  at  $30^\circ$ .

(i) Dipole moment (ii) Torque (iii) Maximum torque

**Given:**  $q = 1 \times 10^{-6} \text{ C}$ ;  $2l = 0.02 \text{ m}$ ;  $E = 10^5 \text{ N C}^{-1}$ ;  $\theta = 30^\circ$

**Formula:**  $p = q \cdot 2l$ ;  $\tau = pE \sin \theta$ ;  $\tau_{\max} = pE$  at  $\theta = 90^\circ$

**Solution:**

(i)

$$p = 1 \times 10^{-6} \times 0.02 = 2 \times 10^{-8} \text{ C m}$$

(ii)

$$\tau = pE \sin 30^\circ = 2 \times 10^{-8} \times 10^5 \times 0.5 = 1 \times 10^{-3} \text{ N m}$$

(iii) Maximum torque at  $\theta = 90^\circ$ :

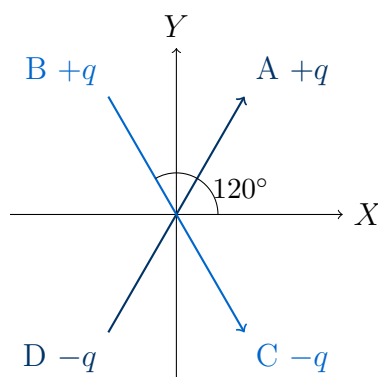
$$\tau_{\max} = pE = 2 \times 10^{-8} \times 10^5 = 2 \times 10^{-3} \text{ N m}$$

**Final Answer**

(i)  $p = 2 \times 10^{-8} \text{ C m}$ ; (ii)  $\tau = 1 \times 10^{-3} \text{ N m}$ ; (iii)  $\tau_{\max} = 2 \times 10^{-3} \text{ N m}$  at  $\theta = 90^\circ$

**Question 27**

Two dipoles AB and CD, each of moment  $p$ , at  $120^\circ$  to each other. Find resultant dipole moment. If  $\vec{E}$  is along  $+X$ , find torque on the system.



Two dipoles  $p$  (AB) and  $p$  (CD) at  $120^\circ$  to each other, with field  $\vec{E}$  along  $+X$

**Solution:****Resultant dipole moment:**

The angle between the two dipole vectors is  $120^\circ$ . Using vector addition:

$$p_{res} = \sqrt{p^2 + p^2 + 2p^2 \cos 120^\circ} = \sqrt{2p^2 + 2p^2(-0.5)} = \sqrt{2p^2 - p^2} = p$$

Direction: bisects the angle, i.e. at  $60^\circ/2 = 30^\circ$  from each dipole. The resultant makes  $30^\circ$  with the  $+X$  axis (as per diagram orientation).

**Torque:**

$$\tau = p_{res} \cdot E \cdot \sin 30^\circ = p \cdot E \cdot \frac{1}{2} = \frac{pE}{2}$$

Direction: along negative  $Z$ -axis (by right-hand rule,  $\vec{p} \times \vec{E}$ ).

**Final Answer**

$$p_{res} = p, \text{ at } 30^\circ \text{ with } +X \text{ axis; } \tau = \frac{pE}{2} \text{ along } -Z \text{ direction}$$

## VI. Electric Flux and Gauss's Law

### Question 28

A point charge of  $1.8 \mu\text{C}$  is at the centre of a cubical Gaussian surface of side 55 cm. Find net electric flux.

**Given:**  $q = 1.8 \times 10^{-6} \text{ C}$ ;  $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

**Formula:**  $\phi = \frac{q_{\text{enc}}}{\epsilon_0}$

**Solution:**

$$\phi = \frac{q}{\epsilon_0} = \frac{1.8 \times 10^{-6}}{8.85 \times 10^{-12}} = 2.034 \times 10^5 \approx 2.0 \times 10^5 \text{ N m}^2\text{C}^{-1}$$

(The size/shape of the Gaussian surface does not matter — only the enclosed charge matters.)

### Final Answer

$$\phi = 2.0 \times 10^5 \text{ N m}^2\text{C}^{-1}$$

**Common Mistake:** The size of the Gaussian surface does NOT affect the total flux — only the charge inside matters.

### Question 29

Net outward flux through a box is  $8.0 \times 10^3 \text{ N m}^2\text{C}^{-1}$ . Find the net charge inside.

**Formula:**  $q_{\text{enc}} = \epsilon_0 \phi$

**Solution:**

$$q_{\text{enc}} = \epsilon_0 \phi = 8.85 \times 10^{-12} \times 8.0 \times 10^3 = 70.8 \times 10^{-9} \text{ C} = 70.8 \text{ nC}$$

### Final Answer

$$q_{\text{enc}} = 70.8 \text{ nC}$$

### Question 30

A point charge produces flux  $-750 \text{ N m}^2\text{C}^{-1}$  through a 10 cm spherical surface.  
(i) Flux if radius doubled? (ii) Magnitude of charge?

**Solution:**

(i) By Gauss's law, flux depends only on enclosed charge, not on radius:

$$\phi_{\text{new}} = -750 \text{ N m}^2\text{C}^{-1}$$

(ii)

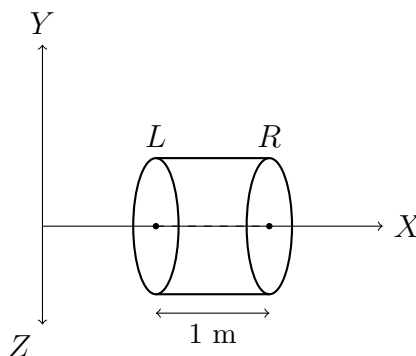
$$q = \varepsilon_0 \phi = 8.85 \times 10^{-12} \times (-750) = -6.6375 \times 10^{-9} \text{ C} \approx -6.64 \times 10^{-9} \text{ C}$$

**Final Answer**(i)  $-750 \text{ N m}^2\text{C}^{-1}$  (unchanged); (ii)  $q = -6.64 \times 10^{-9} \text{ C}$ **Question 31** $\vec{E} = 5 \times 10^3 \hat{i} \text{ N C}^{-1}$ . Find flux through a  $10 \text{ cm} \times 10 \text{ cm}$  square:(i) Plane parallel to YZ-plane; (ii) Plane at  $30^\circ$  to X-axis.**Given:**  $E = 5 \times 10^3 \text{ N C}^{-1}$ ;  $A = (0.1)^2 = 0.01 \text{ m}^2$ **Formula:**  $\phi = EA \cos \theta$ , where  $\theta$  is angle between  $\vec{E}$  and  $\hat{n}$  (normal to surface)**Solution:**(i) Plane parallel to YZ: normal is along  $\hat{i}$ , so  $\theta = 0^\circ$ :

$$\phi = EA \cos 0^\circ = 5 \times 10^3 \times 0.01 \times 1 = 50 \text{ N m}^2\text{C}^{-1}$$

(ii) Plane makes  $30^\circ$  with X-axis: normal makes  $90^\circ - 30^\circ = 60^\circ$  with  $\vec{E}$ :

$$\phi = EA \cos 60^\circ = 5 \times 10^3 \times 0.01 \times 0.5 = 25 \text{ N m}^2\text{C}^{-1}$$

**Final Answer**(i)  $\phi = 50 \text{ N m}^2\text{C}^{-1}$ ; (ii)  $\phi = 25 \text{ N m}^2\text{C}^{-1}$ **Question 32**A hollow cylinder, length 1 m, cross-section  $25 \text{ cm}^2$  placed along X-axis.  $\vec{E} = 50x\hat{i} \text{ N C}^{-1}$ . Find (i) net flux; (ii) enclosed charge.*Hollow cylinder of length 1 m along the X-axis, left face L and right face R*

**Given:**  $L = 1 \text{ m}$ ;  $A = 25 \times 10^{-4} \text{ m}^2$ ; cylinder at  $x = x_1$  to  $x_1 + 1$  (say  $x = 1$  to  $x = 2 \text{ m}$ )

**Formula:**  $\phi = \phi_{\text{right}} - \phi_{\text{left}}$ ;  $q = \epsilon_0 \phi$

**Solution:**

Let cylinder span from  $x = 1$  to  $x = 2 \text{ m}$ :

$$\phi_{\text{right}} = E_{x=2} \cdot A = 50 \times 2 \times 25 \times 10^{-4} = 0.25 \text{ N m}^2\text{C}^{-1}$$

$$\phi_{\text{left}} = E_{x=1} \cdot A = 50 \times 1 \times 25 \times 10^{-4} = 0.125 \text{ N m}^2\text{C}^{-1}$$

$$\phi_{\text{net}} = \phi_{\text{right}} - \phi_{\text{left}} = 0.25 - 0.125 = 0.125 \text{ N m}^2\text{C}^{-1}$$

$$q = \epsilon_0 \phi_{\text{net}} = 8.85 \times 10^{-12} \times 0.125 = 1.106 \times 10^{-12} \text{ C}$$

**Final Answer**

(i)  $\phi_{\text{net}} = 0.125 \text{ N m}^2\text{C}^{-1}$ ; (ii)  $q = 1.106 \times 10^{-12} \text{ C}$

### ★ Likely Exam Question

#### Question 33

An isolated conductor has net charge  $+10 \mu\text{C}$ . A cavity inside contains  $+3.0 \mu\text{C}$ . Find charge on cavity wall and outer surface.

**Solution:**

By Gauss's law, the electric field inside a conductor is zero. Drawing a Gaussian surface inside the conductor material (surrounding the cavity):

$$q_{\text{enc}} = 0 \implies q_{\text{cavity wall}} + 3.0 = 0 \implies q_{\text{cavity wall}} = -3.0 \mu\text{C}$$

Total charge on conductor =  $+10 \mu\text{C}$ . By charge conservation:

$$q_{\text{outer surface}} = (+10) - (-3) = +13 \mu\text{C}$$

**Final Answer**

Cavity wall:  $-3.0 \mu\text{C}$ ; Outer surface:  $+13 \mu\text{C}$

**VII. Electric Intensity due to Linear Charge Distribution****Question 34 (ISC 2006)**

An infinite line charge produces  $E = 9.0 \times 10^4 \text{ N C}^{-1}$  at 2.0 cm. Find linear charge density  $\lambda$ .

**Given:**  $E = 9.0 \times 10^4 \text{ N C}^{-1}$ ;  $r = 2.0 \text{ cm} = 0.02 \text{ m}$ ;  $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

**Formula:**  $E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{2k\lambda}{r}$

**Solution:**

$$\lambda = \frac{E \cdot r}{2k} = \frac{9.0 \times 10^4 \times 0.02}{2 \times 9 \times 10^9} = \frac{1800}{18 \times 10^9} = 10^{-7} \text{ C/m} = 0.1 \mu\text{C/m}$$

**Final Answer**

$$\lambda = 0.1 \mu\text{C/m}$$

★ Likely Exam Question

## VIII. Electric Intensity due to Sheet(s) of Charge

### Question 35

Two metal plates ( $A = 1.0 \text{ m}^2$ ) are 5.0 cm apart with equal and opposite charges. Electric field between them =  $55 \text{ N C}^{-1}$ . Find charges on plates.

**Given:**  $A = 1.0 \text{ m}^2$ ;  $E = 55 \text{ N C}^{-1}$ ;  $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

**Formula:**  $E = \frac{\sigma}{\epsilon_0}$ ;  $Q = \sigma A$

**Solution:**

$$\sigma = \epsilon_0 E = 8.85 \times 10^{-12} \times 55 = 486.75 \times 10^{-12} \text{ C/m}^2$$

$$Q = \sigma A = 486.75 \times 10^{-12} \times 1.0 \approx 4.87 \times 10^{-10} \text{ C} \approx \pm 4.9 \times 10^{-10} \text{ C}$$

### Final Answer

$$Q = \pm 4.9 \times 10^{-10} \text{ C}$$

### Question 36

Two large parallel plates have surface charge density  $\sigma = 17.0 \times 10^{-12} \text{ C m}^{-2}$  (equal and opposite). Find  $E$  to the left, right, and between the plates.

**Given:**  $\sigma = 17.0 \times 10^{-12} \text{ C m}^{-2}$ ;  $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

**Formula:** For two opposite-charged plates:  $E_{\text{between}} = \frac{\sigma}{\epsilon_0}$ ;  $E_{\text{outside}} = 0$

**Solution:**

For a parallel plate capacitor with  $+\sigma$  and  $-\sigma$ :

- Each plate produces  $E = \frac{\sigma}{2\epsilon_0}$ .
- **Between** the plates: fields add  $\Rightarrow E = \frac{\sigma}{\epsilon_0} = \frac{17.0 \times 10^{-12}}{8.85 \times 10^{-12}} \approx 1.92 \approx 2.0 \text{ N C}^{-1}$
- **Outside** (left or right): fields cancel  $\Rightarrow E = 0$

### Final Answer

Left of plates: 0; Right of plates: 0; Between plates:  $\approx 2.0 \text{ N C}^{-1}$

## IX. Charged Spherical Shell

## Question 37

A conducting sphere of radius 1.2 m has surface charge density  $80.0 \mu\text{C}/\text{m}^2$ . Find (a) total charge, (b) total electric flux.

**Given:**  $R = 1.2 \text{ m}$ ;  $\sigma = 80.0 \times 10^{-6} \text{ C}/\text{m}^2$ ;  $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

**Formula:**  $Q = 4\pi R^2\sigma$ ;  $\phi = \frac{Q}{\epsilon_0}$

**Solution:**

$$Q = 4\pi R^2\sigma = 4 \times 3.14159 \times (1.2)^2 \times 80 \times 10^{-6} = 4 \times 3.14159 \times 1.44 \times 80 \times 10^{-6} = 1.4476 \times 10^{-3} \text{ C} \approx 1.45 \text{ mC}$$

$$\phi = \frac{Q}{\epsilon_0} = \frac{1.45 \times 10^{-3}}{8.85 \times 10^{-12}} = 1.638 \times 10^8 \approx 1.64 \times 10^8 \text{ N m}^2\text{C}^{-1}$$

## Final Answer

$$Q \approx 1.45 \times 10^{-3} \text{ C}; \quad \phi \approx 1.64 \times 10^8 \text{ N m}^2\text{C}^{-1}$$

## Question 38

A conducting sphere of radius 10 cm;  $E = 1.5 \times 10^3 \text{ N C}^{-1}$  pointing radially inward at 20 cm from centre. Find charge on sphere.

**Given:**  $R = 0.10 \text{ m}$ ;  $r = 0.20 \text{ m}$ ;  $E = 1.5 \times 10^3 \text{ N C}^{-1}$  (inward)

**Formula:**  $E = k \frac{|q|}{r^2}$

**Solution:**

Inward direction  $\Rightarrow$  charge is **negative**.

$$|q| = \frac{Er^2}{k} = \frac{1.5 \times 10^3 \times (0.2)^2}{9 \times 10^9} = \frac{1.5 \times 10^3 \times 0.04}{9 \times 10^9} = \frac{60}{9 \times 10^9} = 6.67 \times 10^{-9} \text{ C}$$

## Final Answer

$$q = -6.67 \text{ nC (negative, since field points inward)}$$

## Question 39

A metal sphere of radius 12 cm carries charge  $1.6 \times 10^{-7} \text{ C}$  uniformly distributed. Find  $E$  at:

(i) inside the sphere; (ii) just outside the surface; (iii) 18 cm from the centre.

**Given:**  $R = 0.12 \text{ m}$ ;  $Q = 1.6 \times 10^{-7} \text{ C}$ ;  $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$ ;  $k = 9 \times 10^9 \text{ N m}^2\text{C}^{-2}$

**Formula:** (i)  $E_{\text{inside}} = 0$  for a conductor; (ii)  $E = \frac{\sigma}{\epsilon_0} = k \frac{Q}{R^2}$ ; (iii)  $E = k \frac{Q}{r^2}$

**Solution:**

(i) **Inside a conductor:**  $E = 0$ .

(ii) **Just outside the surface ( $r = R = 0.12 \text{ m}$ ):**

$$E = k \frac{Q}{R^2} = \frac{9 \times 10^9 \times 1.6 \times 10^{-7}}{(0.12)^2} = \frac{1440}{0.0144} = 1.0 \times 10^5 \text{ N C}^{-1}$$

(iii) **At  $r = 0.18 \text{ m}$ :**

$$E = k \frac{Q}{r^2} = \frac{9 \times 10^9 \times 1.6 \times 10^{-7}}{(0.18)^2} = \frac{1440}{0.0324} = 4.44 \times 10^4 \approx 4.4 \times 10^4 \text{ N C}^{-1}$$

**Final Answer**

(i)  $E = 0$ ; (ii)  $E = 1.0 \times 10^5 \text{ N C}^{-1}$ ; (iii)  $E = 4.4 \times 10^4 \text{ N C}^{-1}$

**Common Mistake:** For a conducting sphere, charge resides only on the surface. Inside the sphere (even for a solid sphere),  $E = 0$ . The field outside behaves as if all charge is at the centre.